

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: IV

Subject Name: Hydraulics - II

Subject Code: BTCVC401_Y18

Max Marks: 60

Date: 16/01/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

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|--|----------|---|
| A) Differentiate Between open channel and pipe flow. | Remember | 6 |
| B) Enlist flow measuring devices and explain pitot tube. | Remember | 6 |
| C) Find the time required to lower the water level from 4 m to 2 m in reservoir of dimension 75 m X 75 m, by (i) a rectangular notch of length 2.0 m, (ii) a right angled V notch is used. Take $C_d = 0.60$. Take all other data same. | CO 1 | 6 |

Q.2 Solve Any Two of the following.

12

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|---|------------|---|
| A) Define the term most economical section of channel. What are the conditions for the rectangular channel of the best section? | CO 1 | 6 |
| B) Explain Specific energy and Specific energy curve with neat diagram. | Understand | 6 |
| C) An irrigation channel of trapezoidal section, having side slopes 3 horizontal to 2 vertical, is to carry a flow of $10 \text{ m}^3/\text{s}$ on a longitudinal slope of 1 in 5000. The channel is to be lined for which the value of friction coefficient in Manning's formula is $n = 0.012$. Find the dimensions of the most economic section of the channel. | CO 1 | 6 |

Q. 3 Solve Any Two of the following.

12

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| A) State the gradually varied flow assumptions and derive the dynamic equation of gradually varied flow. | CO 2 | 6 |
| B) Define gradually varied flow and rapidly varied flow. Write a note on hydraulic jump. | CO 2 | 6 |
| C) A jet of water 75 mm diameter having a velocity of 20 m/s, strikes normally a flat smooth plate. Determine the thrust on the plate (a) if the plate is at rest, (b) if the plate is moving in same direction as the jet with a velocity of 5 m/s. | CO 3 | 6 |

Also find the work done per second on the plate in each case and efficiency of the jet when plate is moving.

Q.4 Solve Any Two of the following. **12**

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|--|-------------------|----------|
| A) Elaborate major component parts of Pelton wheel, Francis and Kaplan Turbine with neat sketch. | Remember | 6 |
| B) Describe draft tube stating it's need and performance, also state different shapes of it. | Remember | 6 |
| C) Explain working of Centrifugal pump in detail. | Understand | 6 |

Q.5 Solve Any Two of the following. **12**

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|---|-------------|----------|
| A) Explain – Laminar boundary layer, Turbulent boundary layer, Laminar sub layer and Separation of boundary layer. | CO 2 | 6 |
| B) Write a short explanation about following terms: Boundary layer thickness, Displacement thickness, Momentum thickness, Energy thickness and coefficient of Drag. | CO 2 | 6 |
| C) Determine the thickness of a boundary layer at the trailing edge of smooth plate of length 4m and of width 1.5m, when the plate is moving with a velocity of 4 m/s in stationary air. Take kinematic viscosity of air as $1.5 \times 10^{-5} \text{ m}^2/\text{s}$. | CO 2 | 6 |

***** End *****